Noise-Band Intraday Momentum in Nasdaq-100 Futures

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Abstract

This paper adapts the intraday momentum strategy of Zarattini et al. (2024), which trades breaks from a "noise area" around the open on the SPY ETF, to Nasdaq-100 futures (NQ) at a fixed size of one micro contract and under constraints that permit unattended webhook execution. On one-minute data from 2020 to July 2026 the executable rule earns +$10.26 per trading day net of costs (day-block bootstrap 95% interval [+$3.27, +$17.25]), with a daily Sharpe ratio of 1.14 and positive results in each of seven calendar years, two of them only marginally. The paper reports the degradation at each step from the published specification to the executable one, a lookahead audit of the trailing stop, and execution stress tests showing that one index point of slippage on stop fills removes 22% of the edge. A three-year block bootstrap of drawdown gives a 95th-percentile maximum drawdown of $3,406 as backtested and $4,081 under combined lag and slippage, which determines the minimum account size for deployment. The rule is one of two survivors among 144 pre-registered tests on the same archive, all intervals are in-sample, and no live contract has been traded. Pre-registered thresholds for the forward test are stated.

1 Introduction

Zarattini et al. (2024) report a strong risk-adjusted result for an intraday momentum rule on SPY from 2007 to 2024: price is said to be inside a noise area around the open until it has moved further than a trailing average of past intraday moves at the same time of day, and a break outside that area, in the direction of volume-weighted average price, is followed. The claim is a restatement of two mechanisms already documented in the intraday momentum literature: large investors execute over the day rather than at once, so a flow large enough to push price outside its normal range by late morning is often still being executed in the afternoon (Gao et al., 2018); and as the session progresses, option market makers hedging their books buy into rises and sell into falls, adding to a move rather than reversing it (Baltussen et al., 2021).

This paper asks a narrower question than the published one: whether the rule, rebuilt for an instrument, a size, and an execution path that a small systematic trader can actually use, retains an edge after costs. Three changes are made to the published specification. The instrument is NQ futures rather than the S&P ETF. The size is a fixed one micro contract rather than volatility-scaled, because a proprietary-firm account requires a known worst case. And the rule

is rebuilt on five-minute bars with a resting stop order so that a webhook can execute it without a person watching. Each of these costs something, and the paper measures what.

The paper's companion study of a thirty-minute opening range breakout on the same archive (Chuk, 2026) documents the research programme and the 144-test registry within which this rule was found. The two rules are correlated at 0.34 on days both trade.

2 Data

The archive is the same as in the companion paper: Databento GLBX.MDP3 one-minute OHLCV for the NQ parent symbol, 6 June 2010 to 3 July 2026. The continuous front-month series selects, for each session, the contract with the highest volume in the prior session; signals are computed on the continuous series and fills on the tradable contract. The headline period is 2020 to July 2026. The 2014 to 2019 period was checked and was flat, which matches the companion finding that the instrument's intraday behaviour changed around 2020.

The unit of observation is the trading day, not the trade. The rule can take more than one trade in a session and holds positions to a trailing stop, so trades within a day are not independent; expectancy, intervals, and Sharpe ratios are computed from daily profit and loss. Per-trade figures are reported for reference only.

3 Specification

Let t index minutes of the regular session and let σ_t be the average absolute move from the open at minute t over the prior fourteen sessions,

$$\sigma_t = \frac{1}{14} \sum_{k=1}^{14} \left| \frac{P_t^{(k)} - O^{(k)}}{O^{(k)}} \right|,$$

where $P_t^{(k)}$ is the price at minute t on session k sessions prior and $O^{(k)}$ is that session's open. The noise band at minute t is

$$U_t = \max(O, C_{-1})\,(1 + \sigma_t), \qquad L_t = \min(O, C_{-1})\,(1 - \sigma_t),$$

with O the current session's open and C_{-1} the prior close. Table 1 gives the frozen rule.

The conflict lock is not part of the published rule. It exists because the proprietary firms through which the rule would be deployed treat a long in one account and a short in another on the same product as hedging and prohibit it, and because a webhook order from one rule would otherwise close or flip the other's position on the same symbol.

4 Results

4.1 Year by year

Table 2 and Figure 1 report the frozen build at one MNQ. The rule trades on about 60% of sessions, 1.35 trades per session on average, and most trades end on the trailing stop. All seven

Table 1: Frozen specification, the build the paper script runs.

Rule	As frozen
Band	U_t and L_t as above, fourteen-session lookback, recomputed per minute of the day
Entry checks	Only at :00 and :30, from 10:00 to 15:30 ET. Long if the bar close is above U_t and above VWAP; short if below L_t and below VWAP
Fill	Open of the next bar
Stop	Trailing: the band or VWAP, whichever is closer. Recomputed every five minutes at the bar close; active from the next bar
Exit	Stop hit, or flat at the 15:55 bar
Size	One MNQ ($2 per index point)
Costs	$2 per round trip
Conflict lock	No new short while the companion ORB rule holds a long; an open short is closed on the bar ORB enters

years are positive, but 2020 made $49 in total and 2025 made $835; the edge is concentrated in 2021 to 2024 and the first half of 2026. The full-period daily interval is [+$3.27, +$17.25] and the historical maximum drawdown at one contract is $2,458.

Table 2: Frozen build, one MNQ, net of $2 per round trip. Sharpe is from daily P&L, annualised by $\sqrt{252}$.

Year	Sessions	Trades	$/day	$/trade	Sharpe (daily)	Year total
2020	246	331	+0.20	+0.15	0.03	+49
2021	251	314	+7.99	+6.39	1.27	+2,005
2022	250	401	+18.83	+11.74	1.84	+4,707
2023	248	310	+14.15	+11.32	2.25	+3,508
2024	249	322	+13.85	+10.71	1.63	+3,448
2025	247	325	+3.38	+2.57	0.28	+835
2026 (to July)	125	181	+16.23	+11.21	1.16	+2,029
All	1,616	2,184	+10.26	+7.59	1.14	+16,582

4.2 From the published rule to the executable one

Table 3 reports each build between the first test of the published rule on one-minute bars and the frozen executable version. Moving from one-minute to five-minute bars cost about $0.60 per day. Replacing the continuously recomputed stop with a resting order moved every five minutes, which is what a webhook can place, cost a further $1.50. The VWAP-side filter, which requires the entry to be on the VWAP side of the band, recovers a small amount. These are the costs of a version that can be executed unattended, and they are reported so that they are known before, rather than discovered in, a live account.

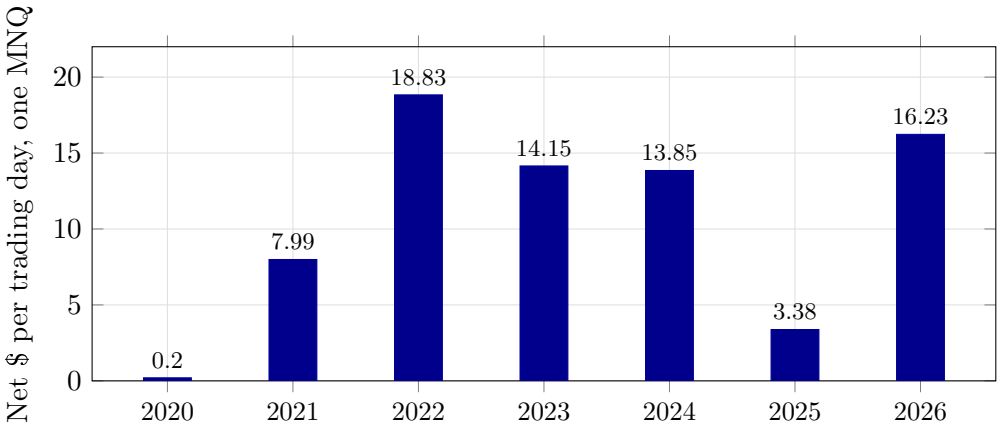


Figure 1: Net dollars per trading day by year, one MNQ, frozen build. 2026 is January to July.

Table 3: Build-by-build results, 2020 to July 2026, one MNQ, per trading day. Day-block bootstrap intervals.

Version	Per day	95% CI	2020–23	2024–26	Note
One-minute bars, original test	+$12.16	[+3.8, +21.3]	+$13.88	+$9.41	Sharpe ∼1.07; positive every year
Five-minute rebuild	+$11.57	[+2.81, +20.73]	+$11.53	+$11.63	+$12.34 per trade
Executable: resting stop, moved every 5 min	+$10.05	[+3.24, +17.03]	+$10.17	+$9.87	The version a webhook can place
With VWAP-side filter (frozen)	+$10.26	[+3.27, +17.25]	+$10.32	+$10.16	Sharpe 1.14; max DD $2,458

5 Lookahead audit

An independent review of the companion study raised one question about this rule: whether the trailing stop could use information from the bar on which it was computed. The code was audited and it cannot. On each five-minute bar the stop is checked first, against the level set at the close of the previous bar. Entries are decided next, on the bar's close. Only then is a new stop level computed from that bar's band and VWAP, and it becomes active on the following bar.

6 Execution stress

The rule exits through its trailing stop on most days, so slippage on stop fills affects it more than it affects a rule with a fixed bracket. Table 4 reports the effect of a one-bar lag in the stop update, one index point of slippage on every stop fill, and both together.

Table 4: Execution stress, one MNQ, 2020 to July 2026, per trading day.

Scenario	Per day	2020–23	2024–26	Sharpe	Max drawdown
As backtested	+$10.26	+$10.32	+$10.16	1.14	$2,458
Stop update one bar late	+$9.52	+$10.49	+$7.97	1.04	$2,932
+1 point slippage on every stop fill	+$7.98	+$8.02	+$7.93	0.88	$2,688
Both	+$7.39	+$8.33	+$5.87	0.80	$3,510

One point of slippage on the stop removes 22% of the edge. Lag and slippage together remove 28% and take the 2024 to 2026 figure to \$5.87 per day. The stressed historical drawdown of \$3,510 is the figure that decides account size, and a single historical path understates the tail.

7 Drawdown and account sizing

Three-year paths were bootstrapped from the daily series in day blocks, 5,000 paths per scenario. Table 5 reports the distribution of maximum drawdown.

Table 5: Three-year maximum drawdown, one MNQ, 5,000 day-block bootstrap paths.

Scenario	Median	95th pct	99th pct	$P(>\$3{,}000)$	$P(>\$4{,}500)$
As backtested	\$1,926	\$3,406	\$4,311	10%	1%
+1 pt stop slippage	\$2,168	\$3,943	\$5,047	18%	2%
Lag and slippage	\$2,246	\$4,081	\$5,272	21%	3%

A \$100,000 proprietary-firm account at the firms considered permits \$3,000 of drawdown; the rule at one MNQ breaches that in 10 to 21% of three-year paths. A \$150,000 account permits \$4,500, breached in 1 to 3%. The rule therefore requires a \$150,000 account at one MNQ, and nothing larger than one contract until the forward record justifies it. This is a sizing conclusion that follows directly from the stress tests and would have been invisible to a backtest that assumed midpoint stop fills.

8 Relation to the opening range breakout

The two surviving rules are correlated at 0.34 on days both trigger: low enough to be worth holding together, high enough that they cannot be treated as independent. Both prefer trending mornings and both lose on reversal days. A combined script netting the two positions in one account does not yet exist. Until it does the rules must run on separate accounts, for two mechanical reasons (a webhook order from one rule would close or flip the other's position on the same symbol; the ORB 14:00 close message would flatten a noise-band position meant to run to 15:55) and one contractual one (the firms prohibit opposing positions across accounts on the same product, hence the conflict lock of Section 3).

In a joint block bootstrap of the two rules on a funded 50K account, adding one MNQ of this rule to the companion rule at its \$400 cap roughly doubled median gross payouts over three years while raising the three-year probability of breaching the account from 31% to 39%. That result is exploratory and conditional on both backtest edges being real.

9 Forward test protocol

The rule has been in paper-forward testing since 28 September 2026 through a script on the same chart as the companion rule, logging each trade without sending orders. Four measurements are recorded for every stop event: the stop price the script sent, the actual fill, the time from alert to broker acknowledgement, and the market price at acknowledgement, so that slippage from

the fill can be separated from slippage from the delay. The average, 90th, and 95th percentile stop slippage are tracked, because a 0.4-point average can hide occasional four- to eight-point gaps.

The following thresholds were written before the forward test began and will not be moved.

- After about 60 forward sessions: average stop slippage below 0.5 points means the backtest holds; 0.5 to 1.0 points means the edge is real but about a quarter smaller; above 1.0 points means re-underwriting before any account is purchased. Frequent multi-point outliers are a larger warning than any average.
- At 100 forward sessions: the rule continues only if the forward daily average is above zero and the exit mix resembles the backtest.
- Real capital only if the forward daily interval excludes zero on its own, without reference to the backtest.

10 Limitations

1. **In-sample.** The rule was one of 144 tested on the same archive; a merely positive forward result is insufficient, and the honest out-of-sample expectancy is lower than the printed figure by an amount that cannot be estimated from these data.
2. **Two near-flat years.** 2020 and 2025 contributed \$49 and \$835 respectively. The edge is concentrated in 2021 to 2024 and the first half of 2026, and a reader who weights recent data more heavily should note that the stressed 2024 to 2026 figure is \$5.87 per day.
3. **Execution dominates.** The edge is about \$10 per day. Realistic trailing-stop fills plus webhook delay consume a large share of it, which is why the forward test is designed around measuring slippage rather than profit.
4. **The published result is not reproduced.** The SPY result of Zarattini et al. (2024) is theirs; this paper reports only what the adapted rule does on NQ in this archive. The simplest version of the first-half-hour predictability of Gao et al. (2018) was tested directly on NQ in the companion registry and did not survive costs, so the published literature is treated as a reason the mechanism is plausible, not as evidence for these rules.
5. **Specification correction.** An earlier draft stated that entries were checked every five minutes. Entries are checked at :00 and :30 only; the stop is what updates every five minutes. The code was always as described in Table 1; the prose was not.

11 Conclusion

A noise-band intraday momentum rule adapted from the SPY literature to NQ futures, at a fixed one micro contract and under constraints permitting unattended execution, earns about \$10 per trading day net of costs over six and a half years with a daily Sharpe ratio of 1.14, positive in every year and marginal in two. The adaptation costs are measured and small; the

execution costs are measured and large, with one point of stop slippage removing roughly a fifth of the edge. Drawdown bootstraps under stress set the minimum account at $150,000 for one contract. The rule has not traded a live contract and will not until the pre-registered forward thresholds of Section 9 are met.

References

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